AUTOMATIC BIKE SIDE STAND SLIDER

MINI PROJECT REPORT

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of

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DECLARATION

We undersigned hereby declare that the report "**AUTOMATIC BIKE SIDE STAND SLIDER**", submitted for the partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a Bonafide work done by us under supervision of Mrs. FATHIMA SHERIN M K. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources.

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CERTIFICATE

This is to certify that this report entitled **“AUTOMATIC BIKE SIDE STAND SLIDER”** submitted by ADWAITH KS(CCV22MC003), AFTHAB ASHRAF(CCV22MC004), MOHAMMED ROSHAN NP(CCV22MC010), NIHAS PK(CCV22MC012) to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements of the award of the Degree of Bachelor of Technology in Mechatronics Engineering is a Bonafide record of the work carried out by them under our guidance and supervision. This report is any form has not been submitted to any other University or Institution for any purpose.

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ABSTRACT

The Automatic Motorcycle Side Stand Retraction System is designed to enhance rider safety by preventing accidents that may occur due to a forgotten or improperly retracted side stand. This system employs a Hall-effect speed sensor to continuously monitor the speed of the motorcycle and a microcontroller (Arduino) to process the speed data and control the movement of the side stand. The motor control is achieved using an L298N motor driver, which regulates a Brushed DC motor responsible for retracting the stand.

When the motorcycle's speed exceeds 10 km/h, the microcontroller sends a signal to the motor driver, activating the Brushed DC motor to retract the side stand automatically. To ensure precise operation and prevent overextension or excessive retraction, the system incorporates two limit switches that act as position sensors. These switches detect when the stand is fully extended or retracted, stopping the motor at the correct position and preventing mechanical strain or misalignment.

The entire system is powered by the motorcycle's 12V battery, ensuring seamless integration without the need for an additional power source. By utilizing cost-effective and readily available components, this system offers an affordable yet highly effective solution to a common rider safety issue. Additionally, it requires only minimal modifications to the motorcycle, making it a practical enhancement for various models. By automating the side stand retraction process, this system significantly reduces the risk of accidents, improves riding convenience, and enhances overall road safety.

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LIST OF ABBREVIATIONS

1.	AC	Alternating Current
2.	AI	Artificial Intelligence
3.	ABS	Anti-lock Braking System
4.	ARAS	Advanced Rider Assistance System
5.	AVR	Automatic Voltage Regulator
6.	BDC	Brushed Direct Current
7.	BLDC	Brushless Direct Current
8.	C	Common
9.	DC	Direct Current
10.	DIY	do-it-yourself
11.	DSP	Digital Signal Processor
12.	HVAC	Heating, Ventilation and Air Conditioning
13.	IC	Integrated Circuit
14.	IDE	Integrated Development Environment
15.	IMU	Inertial Measurement Unit
16.	IoT	Internet of Things
17.	LED	Light Emitting Diode
18.	NC	Normally Closed
19.	NO	Normally Open
20.	PWM	Pulse Width Modulation
21.	RC	Radio Controlled
22.	RISC	Reduced Instruction Set Computer
23.	RPM	Revolutions Per Minute
24.	UNO	Universal Numbering Operation
25.	USB	Universal Serial Bus
26.	VFD	Variable Frequency Drive

CHAPTER 1
INTRODUCTION

This chapter deals with the introduction and objectives of the project "Automatic Bike Side Stand Slider".

1.1 INTRODUCTION

Two-wheelers have become one of the most widely used means of transportation due to their affordability, fuel efficiency, and ability to navigate through traffic with ease. However, with the increase in two-wheeler usage comes the responsibility to ensure rider safety. One common but often overlooked issue is the rider forgetting to retract the side stand before moving. This seemingly minor oversight can lead to serious accidents especially during turns or on uneven roads when the extended stand catches on the ground and destabilizes the vehicle.

The Automatic Bike Side Stand Slider project is designed to address this exact problem. The core idea is to automate the retraction of the bike's side stand when the vehicle starts moving. Using a speed sensor, the system monitors the bike's motion, and once the speed exceeds a certain threshold 10 km/h a microcontroller processes this data. If the stand position sensor detects that the stand is still down, the system activates an actuator or motorized mechanism to retract it automatically. This removes the dependency on human memory and greatly reduces the risk of accidents caused by a forgotten side stand.

The system architecture includes essential components such as a speed sensor, a side stand position switch, a microcontroller unit (like an Arduino), and an electromechanical actuator. The components work in coordination to ensure that the stand is retracted only when necessary—avoiding false triggers and preserving energy. The system is also designed to be cost-effective, compact, and easily adaptable, making it suitable for a wide variety of two-wheeler models.

This project is particularly beneficial for beginner riders, senior citizens, and delivery personnel who frequently start and stop their bikes. By automating a commonly neglected aspect of bike safety, this system enhances rider confidence and overall road safety. Moreover, it showcases the practical use of embedded systems and sensors to solve real-world challenges in transportation.

1.2. OBJECTIVES

The primary objective of this project is to design and implement an automated system that eliminates the safety hazard caused by a motorcycle's side stand being left extended while the vehicle is in motion. This issue, though simple, can lead to dangerous accidents, particularly during sharp turns or while navigating uneven surfaces. The proposed solution involves using a speed sensor to monitor the motion of the bike and a microcontroller-based system to automatically retract the side stand when a preset speed threshold 10 km/h is exceeded. This automation ensures that the stand is retracted without any manual input from the rider, thereby reducing the risk of human error. Another key objective is to enhance road safety by introducing a preventive mechanism that addresses one of the most common causes of two-wheeler mishaps. The system is designed to be cost-effective, energy-efficient, and reliable, making it suitable for real-world applications and easy to retrofit onto existing motorcycles. By doing so, the project aims to make the solution accessible to a wide range of users, including daily commuters, delivery drivers, and new or elderly riders who may be more prone to forgetting the sidestand.

CHAPTER 2

LITERATURE REVIEW

This chapter deals with the literature survey of the project.

Jamal Uddin Ahmed, S.M Minhazul Islam, Avishek Das, Moin Uddin Ahmed Babar. Development and fabrication of automated bike stand retrieval system. 2021:[1] The growing concern over road safety, especially among two-wheeler riders, has led researchers to explore automated systems that can prevent common causes of accidents. One such hazard is riding with the side stand engaged, which can cause the rider to lose balance during a turn or while accelerating. As a result, various studies have focused on automated bike stand systems to address this issue. Previous research efforts have introduced electromechanical systems that automatically retract the side stand once the vehicle starts or reaches a certain speed. These systems typically use components such as limit switches, sensors, and DC motors to detect the stand's position and actuate its movement. Some designs implement microcontrollers or relays to control the stand based on ignition status or gear shift. For example, mechanisms have been proposed where the stand retracts only when the vehicle is in gear, preventing the bike from being ridden with the stand deployed. Other approaches include the use of tilt sensors or speed sensors to trigger retraction when the bike starts moving. The reviewed paper, titled “Development and Fabrication of Automated Bike Stand Retrieval System”, presents a practical implementation of such a mechanism. It focuses on the design and fabrication of an automated system that retracts the side stand once the bike reaches a certain speed, using a combination of sensors and mechanical actuators. This innovation directly addresses the common issue of riders forgetting to retract the stand, which often leads to serious accidents, especially while turning. Compared to earlier models that relied solely on ignition input or manual switches, this system improves safety by integrating real-time sensing and automatic actuation, ensuring the stand retracts precisely when needed.

Manisha, Hemalgi Patil, Avadhoot Patil, Pushkraj Patil, Mihir Patil, Arnav Patil. Automatic motorbike stand slider using Arduino UNO. 2024:[2] Even though two-wheelers are a widely known form of transportation, their weaker structure makes them more prone to accidents. These collisions are caused by a number of things, including going over speed limits, failure to obey traffic rules, and forgetting to raise the side stand. When starting their bikes, riders keep forgetting to raise their side stands, which is one of the most prevalent reasons for these accidents. It suggests an automated side-stand system that, upon starting the two-wheeler, automatically raises the stand in order to tackle this issue. For this project, they have created a demonstration model with a side-stand-attached frame that is powered by a servo motor and a mock bike starter. This project introduces a fully automated and also a manually operated side-stand mechanism for motorcycles, which improve. In case the ignition switch of this system is not working for these, it also has a push-button by which a person can manually control the stand. This system also has LEDs that indicate the status of the stand slider. If the LED is on, which indicates the stand-slider is not lifted, it will be only off when the stand slider is lifted, which is very useful for manually controlling the stand slider. It works as a double safety mechanism. The design of this project is an automatic motorcycle stand system, which involves some very important points. And those are: • Servo Motor: A servo motor with high torque has been used to support the weight and reduce the mechanical strains caused by the motorcycle's stand in progress of extension and retractions to our calves. • Microcontroller: (for instance, Arduino UNO) This component (a microcontroller) works as a central unit to control the received inputs from the push buttons and also to manage the servo motor's operations and illuminate the LED to indicate the status of the motorcycle's stand. • LED: Work as an indicator that signifies the status of the stand as either deployed or retracted. • Ignition switch: It is helpful for making the stand slider fully automated. It will lift the stand when the bike's key is inserted in the ignition switch for starting the vehicle. • Push Button: It is basically used to manually control the stand in case the ignition switch is not working. • Power Supply: The system is usually powered by a 12V battery, as it will provide the power needed by every component of the stand slider

K.Kumar, P..S Aravind, N.Narasimha. V.Dhanunjaya,T.Chaithanya. Automatic motorbike side stand slider. 2023:[3]In this modern developing world, automobiles play a very important role especially bikes. These are mainly used for travelling from one place to another. while riding the bike the rider may get accident by number of reasons. In that one of the main and important reason is due to not lift the side stand in properly. To overcome this, we have designed a “AUTOMATIC MOTOR BIKE SIDE STAND SLIDER”. The side stand will get in the position of horizontal at the time of ignition functionality. In this regard, the operation work will be followed in between ignition side stand. This automation will be followed by electronic devices with Side stand, Motor, Electronic circuit board. Components used: An ignition switch, starter switch or start switch is a switch in the control system of a motor vehicle that activates the main electrical systems for the vehicle, including "accessories". The ignition switch performs a primary function in that it connects the circuit that provides voltage to your starter motor, allowing the engine to crank over and the engine to eventually start. FRAME 1. It is a basic component of the project 2. It holds the component like servo motor and stand to lift. 3.It is made of metal and the edges are welded together MOTOR DRIVER L293D is a dual H-bridge motor driver integrated circuit (IC). Motor drivers act as current amplifiers since they take a low-current control signal and provide a higher-current signal. This higher current signal is used to drive the motors. ATMEGA 328 1. ATmega [328] is an 8-bit, 28-Pin AVR Microcontroller, manufactured by Microchip, follows RISC Architecture and has a flash-type program memory of 32KB. 2.ATmega-328 is an AVR Microcontroller having twenty-eight (28) pins in total. SERVO MOTOR MG995 Metal Gear Servo Motor is a high-speed standard servo can rotate approximately 180 degrees (60 in each direction) used for airplane, helicopter, RC-cars and many RC model. It is a Digital Servo Motor which receives and processes PWM signal faster and better. It equips sophisticated internal circuitry that provides good torque, holding power, and faster updates in response to external forces. WIRE DESCRIPTION: 1.RED – Positive 2. Brown – Negative 3. Orange – Signal

Harshal Butle, Mohit Waghade, Shubham Govindwar. Auto retreat side stand. 2022:[4]The Auto Retreat Side Stand system is designed to reduce the possibility of accidents caused by forgetting to retract the side stand while riding a two-wheeler, by automatically lifting the stand when the vehicle starts moving. If the rider forgets to retract the side stand before riding, rider and the engine is suddenly stops in mid of the road then the chance of stand hitting the ground and affecting and in mid of the traffic, this again put the driver in a panic the riders control during the turn increases. In this paper situation. To avoid such risks, we have designed Auto the system of Auto Retreat Side Stand is designed to reduce Retreat Side Stand.The presented mechanism consists of 12 V geared motor powered by motorcycles battery. The system is comprised of spring-loaded piston. The project is using sensors, microcontroller and an actuator as a 12V DC geared motor to control the mechanisms of automatic retreat of the side stand system. The sensor will constantly monitor the rotation of wheel. The main control unit is connected with the sensor to receive signal. The control unit is also connected to the actuator which is connected to side stand assembly. Initially when the wheel is in the rest position, the side stand is in the deployed downward position. At this time the reed switch will not generate any pulses. The pulse generation of reed switch is dependent on rotation of wheel as magnet is attached with the wheel arrangement. When the wheel starts to rotate the magnet also rotate along with it. As the magnetic field passes near the sensor it generate a pulse and sends it to the control unit. The control unit receives the signal from sensor. The controller will send the signal to relay based on the pulses received from sensor. The relay closes the circuit for the DC geared motor. The cam attached to the shaft of motor lifts the follower which is pivoted to the cylinder by a pin. The follower also acts as a locking arrangement to the piston. Initially the piston is in locked position and the spring inside the cylinder is in the compressed position. As the follower rotates, it unlocks the piston. The spring forces the piston and piston further forces the metal strip attached to the modified stand. This action will result in the lifting of the stand upward. The tension spring will hold the side stand in its up position

Sachin J. Ugale, Yogesh R. Pawar, Harshal B. Deore, Saurabh D. Patil. Automatic bike stand for two -wheelers. 2022:[5]While the two-wheelers are concerned, accidents occur due to riding the vehicle in high speed, ignores to use helmets, does not maintain the speed limit and forgets to lift the side stand while riding the vehicles. These are the major source for accidents, forgetting to lift the side stand causes huge accidents. We have developed a new type of side stand which is automatically retracting the side stand through some mechanical and electronic arrangement and it also have some advance feature. In this system microcontroller, Displacement, Vibration Sensor with a dc battery is used. Through the sensor, sensor sense the rotation of the key from OFF position to ON and sends the signal to the microcontroller which is actuate the dc motor which is caused the disengage the stand from the road. The starter consists of a microcontroller circuit used to monitor the starter and then operate the stand sliding mechanism. The stand consists of a motorized system used to operate the stand. The circuit monitors the starter, on starting the bike the side stand is operated by the motor using a shaft to slide from a vertical position to a horizontal position. On turning off the key in other direction to lock bike the system moves the motorized stand shaft in opposite direction so as to move the stand in a direction perpendicular to the bottom frame rod which rests the motor bike on side stand. Thus, we have a fully automated side stand system for motor bikes. It by an observation that the side stand retrieve by mechatronic based circuit is better than mechanical method of stand retrieving It can reduce risk of accidents caused by un removal of side stand definitely this system could be used in all type two wheelers this project could be a revolution in automobile industry because it provides a solution for the problem we are facing in our daily life Running a bike with side stand in its uplift may creates problems, but with the help of our project solve this problem

Raj Mahour,BhupendraKumar Pandey. Automatic Motor Bike Stand Slider. 2022:[6]The Automatic Bike Stand slider can reduce accidents caused by negligence by automatically lifting the side stand when the motorcycle is started. A reduction gear box is installed along with the rotor of the motor, it is inside the motor itself, we do not see it from outside. All of us always install lithium-ion batteries in our motor bikes. This automatic motor bike stand slider method consumes very little power. Therefore, batteries are mainly used extensively in two wheelers. This method mainly works on 12V. First of all, we will turn on the ignition key and after turning on the ignition latch and watch the input to the arduino and then arduino will accept the input as the arduino will get the output. As soon as the arduino is out, the arduino will output to the servo motor as an input. When the rider puts the key in the vehicle, turns the ignition off or on, the microcontroller sends a signal to the arduino by the ignition and receives the signal as an output to indicate whether the process is on or off as soon as the motorist gives the input. If so, microcontrollers receive the signal and send it to the single straight servo motor. The servo motor is connected to the microcontroller and sends the signal. After receiving the signal from the motor driver, the servo motor is fully activated, then there are signals in the servo motor, it tells that the soft of the servo motor is to be rotated in any direction or in which degree the side stand is attached to the servo motor. Then the servo motor rotates according to the signal received from the driver. The results of the study show that the Automatic Bike Stand slider can automatically lift the side stand when the motorcycle is started and return it to its original position when the ignition is turned off.

Shubham Jichkar. Deepesh Kumar, Rushikesh Dhawale, Hrithik Dashmukh. Auotmatic side stand slider assembly. 2020:[7]The project uses a combination of electronic components, including a servo motor, motor driver, and Arduino microcontroller, to automate the side-stand retrieval process. The system is powered by a 12V battery and consumes very low power. The methods used in the study include designing and developing an automatic side stand retrieval system using electronic components, such as a servo motor, motor driver, and microcontroller. The proposed system has practical applications in the automotive industry, particularly in the development of safer and more efficient two-wheeler vehicles. The system can also be used in other applications, such as in the development of autonomous vehicles. The practical applications of the study include reducing the risk of accidents caused by forgotten side stand retrieval, providing an aesthetic accessory for the two-wheeler, and potentially reducing the number of injuries and fatalities related to two-wheeler accidents. The system works on electronic circuits in the bike with the help of electronic components such as Servo motor, motor driver, Arduino for the retrieval of stand and to apply it. The frame is used to mount bike upright using frame. The starter consists of a circuit used to monitor the starter and then operate the stand sliding mechanism. The stand consists of a motorized system used to operate microcontroller the stand. The circuit monitors the starter, on starting the bike the side-stand is operated by the motor using a shaft to slide from a vertical position to a horizontal position. On turning off the key in other direction to lock bike the system moves the motorized stand shaft in opposite direction so as to move the stand in a direction perpendicular to the bottom frame rod which rests the motor bike on side stand. Thus, we have a fully automated side-stand system for motor bikes. The proposed "AUTOMATIC SIDE-STAND SLIDER ASSEMBLY" is a practical solution to minimize accidents caused by riders forgetting to retrieve the side-stand. The system is automated, efficient, and can be used on any type of two-wheeler. The conclusions of the study are that the automatic side stand retrieval system is a necessity for the safety of the rider, reducing the risk of accidents caused by forgotten side stand retrieval, and providing an aesthetic accessory for the two-wheeler.

Sushant.S, Jadhav Malhar.A, Takalkar Atul. Development of side stand retrieval mechanism for two- wheeler. 2020:[8]The study includes the development of a side-stand retrieval mechanism that can automatically lift the side stand when the vehicle is in motion, reducing the risk of accidents caused by forgetting to lift the side stand. The study also found that the implementation of the proposed mechanism can be utilized in heavy two-wheeler vehicles used for carrying heavy loads, and can ensure the safety of new drivers who may forget to retrieve the stand. The constructional feature for the proposed mechanism includes a sprocket situated in the transmission train in between the final drive sprocket and rear wheel sprocket along with a lever mounted on the periphery of the sprocket. When the vehicle is in the parked condition, the lever welded to the side stand rests over the sprocket lever. This condition will prevail whenever the vehicle is rested on the side stand. The rotation of the sprocket will cause the lever to push the side stand lever in an upward direction, and the rest of the work required to retrieve the side stand is done by the spring tension present in the side stand. The conclusions of the study include the development of a side-stand retrieval mechanism that can automatically lift the side stand when the vehicle is in motion, and the potential of the proposed mechanism to reduce the risk of accidents caused by forgetting to lift the side stand. The study also concludes that the implementation of the proposed mechanism can provide a safe and efficient solution for two-wheeler riders, particularly new drivers who may forget to retrieve the stand.

J. Joy, P.G.S.G Premathilaka, K.G.L Dinakara, H.M.T.D Herath. Hardware implementation of automatic side stand for motor bike. 2020:[9] The side stand of motor bike plays an important role while the vehicle is in rest position. But it has some disadvantages when the vehicle is moving, if it is not lifted. If the bike rider forgets to release the side stand, it will cause to accidents when taking left turn. Even though some motor bikes have indication of side stand position in head panel, quite a number of accidents are still occurring due to this particular reason. So, it is necessary to apply a mechanism to lift the side stand even if the rider forgets to lift it when he starts riding. The system developed here is automatically lifting and retracting the side stand through mechanical cum electronic arrangement. The system is working with the sensors, actuators and using the limit switch signals. Mini Arduino board is used to give conditions to control the system and to program it. If the ignition switch turned ON, the input signal is given to the Arduino pro mini by the 10K resistor. According to the Arduino coding, the input signals processed and output signal will be given to the motor control driver. The motor control driver can ON/OFF the motor & change the rotational direction of the motor shaft base on whether to lifting up or retrieving the side stand. The side stand is programmed to be lifted up once the motor bike key turn ON. If the side stand comes to the upper limit, the upper limit switch sends the signal to the Arduino pro mini. According to the coding and command, the motor stop working and further lifting the stand is prevented. If the ignition switch is turned OFF, the side-stand retrieve to bottom level and engage with lower limit switch. In the similar way, after engaging with limit switch, the motor is turned OFF automatically and further movement of side-stand is prevented. If there is any hindrance on the way of lifting-up or retrieving the side-stand, the motor is turned OFF automatically and the buzzer rings an alarm. Current sensor is used in this aspect. Current range is given to the motor based on the Arduino coding. More power is required to lift the side stand, if there is any hindrance on the way of lifting-up or retrieving down. More power means, it needs more voltage and amperes to complete that certain task. If this current exceeds the prescribed limit based on the coding, the motor is turned OFF automatically and the buzzer rings an alarm and indicates the user by LED light. Since 7805 regulators can act as a step-down transformer, it is used to convert 12V motor bike battery voltage into 5V system operating voltage. Excess voltage may damage the system. Once the 7805 regulator is fixed, the power in put wire can be connected directly to the battery. So, there won’t be any changes in the electrical system by installing this new device with existing system. Safety of the new attachment is another concern. The new system must be water resistive. Since the side stand is in bottom part of the motor bike, there is high possibility of getting wet in rainy days. It is obvious that the system will not work properly under such situation. All the electronic accessories are placed in a separate box and kept under the seat to eliminate the above-mentioned problem. If the system failed to work because of any problem, it would be indicated by an indicator light and an alarm sound. The system needs to be reset under such circumstance. Turn off and turn on the start key within few seconds is how the reset is programmed.

Shantanu S.Chilgar, Shubham S. Chilgar. Design and modification of side stand lifting mechanism.2019:[10]The objective of the study is to develop a mechanism to automatically lift the side stand of a two-wheeler when the rider forgets to do so, thereby increasing safety standards and reducing accidents. The study aims to provide a solution to the safety issues related to the side stand and to reduce the number of accidents caused by the side stand. The method used to develop the mechanism involves welding a stainless-steel washer below the gear lever of the bike, and attaching a metal plate to the side stand with an arc welding. A small hole is drilled on the metal plate, and a metal wire is attached to the plate and the washer, which lifts the side stand when the rider shifts gear. When rider starts the bike and forgot to lift side stand, at this moment rider will downshift to engage the gear. Now, the constructed mechanism come into action as rider downshift to engage gear, the mechanism quickly lifts the side stand as rider shifts gear. With the help of thin wire, this system works quickly. It pulls the side stand as rider shifts a gear. This mechanism does not require any type of outer source of energy, it’s a fully mechanical system. The conclusions of the study are that the proposed mechanism is an effective solution to the safety issues related to the side stand, and that it can be applied to all types of vehicles to increase safety standards. The study also concludes that the distribution of road accidental deaths and injuries varies according to age, gender, month, and time, and that males are more vulnerable to accidents than females.

CHAPTER 3

SYSTEM DESCRIPTION

This chapter deals with the description of the system. Basic block diagram=m and basic mechanism. About the software and hardware components and the working automatic bike side stand slider

3.1 BLOCK DIAGRAM

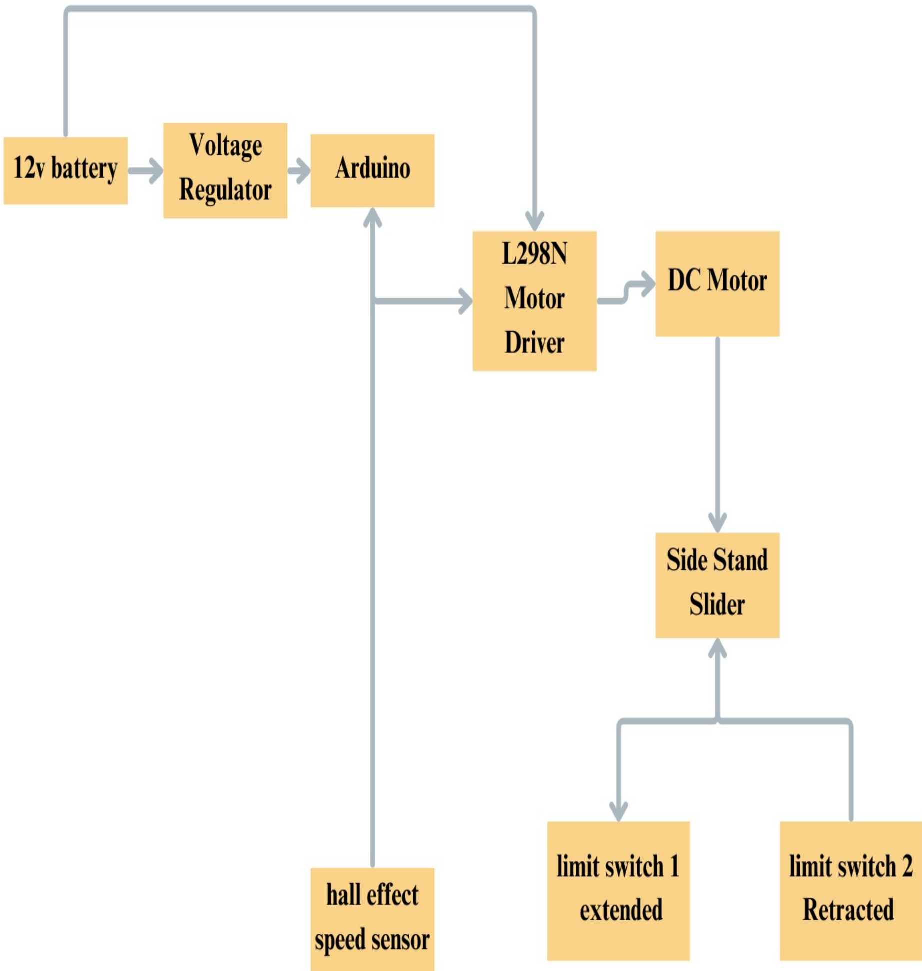

Fig.3.1 Block Diagram

3.2 BASIC WORKING

This is a safety feature for motorcycles that automatically retracts the side stand once the motorcycle reaches a certain speed. The goal is to ensure that the stand doesn't remain down while the bike is moving, preventing it from causing accidents or damage to the motorcycle.

1. Speed Detection

The system starts by continuously monitoring the motorcycle's speed using a speed sensor. This sensor can be a hall effect sensor that detects the rotational speed of the wheel. The sensor sends data to the microcontroller, which is the brain of the system.

2. Microcontroller Processing

Once the sensor sends the speed data, the microcontroller processes the data. The microcontroller is a small, programmable device (like an Arduino, Raspberry Pi, or a custom-made controller) that makes decisions based on the data it receives.

- Logic: When the motorcycle's speed reaches 10 km/h, the microcontroller recognizes this as the threshold for the system to act. It sends a signal to activate the next step.
- Thresholding: For example, when the microcontroller detects 10 km/h, it compares it to the programmed threshold and checks whether the stand is still deployed. If the stand is down, it sends a signal to retract it.

3. Motor Activation

Once the microcontroller decides that the motorcycle has reached the desired speed, it sends a signal to a DC gear motor (a type of motor that allows for controlled motion with reduced size and weight). This motor is responsible for physically retracting the side stand.

- The motor has a mechanical linkage that connects to the side stand. When activated, the motor retracts the stand automatically.
- The gear mechanism ensures that the stand retracts smoothly without overshooting, so it's safely out of the way when the bike is moving.

4. Safe Operation

Once the side stand is retracted, the system ensures that the stand remains in the retracted position while the motorcycle is moving. The side stand should only be deployed when the bike is stationary.

- Safety Mechanism: The system might also incorporate fail-safes to ensure the stand doesn't accidentally deploy while riding. For example, the microcontroller could constantly check the stand's position via a feedback sensor to ensure it stays retracted.
- Manual Override: While the system is automatic, the rider can still manually deploy the stand if needed. The system ensures the side stand is not locked in the retracted position permanently.

3.3 HARDWARE USED

3.3.1 Arduino Uno R3

The Arduino UNO R3 is a popular microcontroller development board that uses the ATmega328P microchip to control electronic devices. It features 14 digital input/output pins, 6 analog inputs, and a USB connection for programming and power supply. The board operates at a clock speed of 16 MHz and can be powered via USB or an external power source. It is programmed through the Arduino Integrated Development Environment (IDE) using C/C++ programming language. The board includes a variety of features like PWM output for controlling devices like motors and LEDs, as well as serial communication for interfacing with other systems. With its user-friendly design and vast community support, the Arduino UNO R3 is widely used in a range of projects, from robotics and home automation to IoT and prototyping. Its open-source nature makes it accessible for beginners while offering enough flexibility for more advanced users.

Fig.3.2 Arduino Uno R3

3.3.2 Voltage Regulator

A voltage regulator is an electronic circuit or device designed to automatically maintain a constant voltage level to power electronic components, ensuring that they receive the correct voltage for optimal performance. It adjusts varying input voltages to produce a stable output voltage, compensating for fluctuations in the supply voltage or changes in the load current. Voltage regulators are essential in preventing over-voltage or under-voltage conditions that could damage sensitive components. Commonly used in power supplies, battery-powered devices, and consumer electronics, voltage regulators play a critical role in ensuring the reliability and longevity of electrical systems.

Fig.3.3Voltage Regulator

3.3.3 Brushed DC motor

A Brushed DC Motor (BDC Motor) is a type of electric motor that converts electrical energy into mechanical motion using direct current (DC) and a simple electromechanical system. It consists of a stator (which provides a stationary magnetic field using permanent magnets or electromagnets), a rotor (armature) with wound copper coils, a commutator that reverses current direction, and carbon brushes that maintain electrical contact with the rotating commutator. When current flows through the armature windings, a magnetic field is generated, interacting with the stator's field to produce rotational motion. The commutator periodically reverses the current in the windings, ensuring continuous rotation in a specific direction. The speed of a brushed DC motor can be controlled by varying the applied voltage, while the direction can be reversed by changing the polarity of the supply.

Fig.3.4 Brushed DC motor

3.3.4 Motor Control Drive

A Motor Control Drive is an electronic system used to regulate the speed, torque, and direction of an electric motor by adjusting the input voltage, current, or frequency. It acts as an interface between the power supply and the motor, ensuring smooth operation, efficiency, and protection from electrical faults such as over current, overheating, or voltage fluctuations. Motor drives can be categorized into AC drives (Variable Frequency Drives - VFDs), DC drives, and servo drives, depending on the type of motor being controlled. In a DC motor control drive, techniques such as Pulse Width Modulation (PWM) are used to control the voltage and speed, while in AC motor drives, frequency and phase control are applied to adjust performance. Advanced motor drives integrate microcontrollers or digital signal processors (DSPs) for precise feedback control, enabling applications in robotics, industrial automation, electric vehicles, and HVAC systems. Modern drives often include sensor-based or sensor less feedback mechanisms, such as Hall effect sensors, encoders, or current sensors, to optimize motor performance, reduce energy consumption, and improve operational reliability.

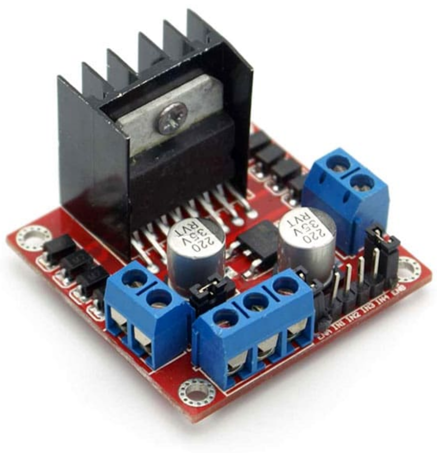

Fig.3.5Motor Control Drive

3.3.5 Hall Effect Speed Sensor

A Hall Effect Speed Sensor is an electronic device that detects changes in a magnetic field to measure the speed or position of a rotating object, such as a motor shaft, wheel, or gear. It operates based on the Hall Effect principle, where a voltage difference is generated when a magnetic field is applied perpendicular to the current flow in a semiconductor material. The sensor typically consists of a Hall element, an LM393 comparator IC for signal processing, and a potentiometer for sensitivity adjustment. When a magnet or a ferromagnetic object (such as a gear tooth) passes near the sensor, the magnetic field variation triggers a voltage change, producing a digital pulse output (High/Low) that corresponds to the rotational speed. By counting these pulses over time using a microcontroller (such as an Arduino), the RPM (Revolutions Per Minute) of the rotating object can be calculated. These sensors are widely used in automotive speedometers, industrial tachometers, motor controllers, and robotics for precise speed and position measurement due to their non-contact operation, high accuracy, and durability in various environmental conditions.

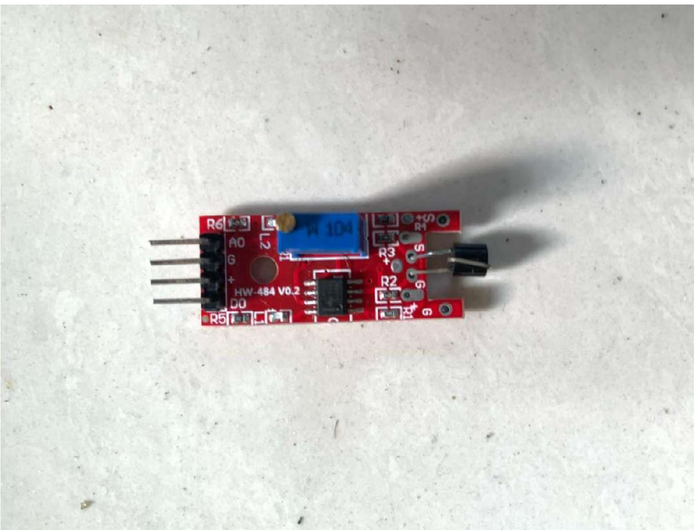

Fig.3.6Hall Effect Speed Sensor

3.3.6 Mini Motor

This motor widely used in low-power applications such as toys, small robots, fans, electric toothbrushes, and DIY electronics projects. This type of motor operates on direct current (DC) and uses brushes and a commutator to transfer electrical energy to the rotor windings, generating motion through electromagnetic interaction. The metallic cylindrical housing contains the rotor, stator magnets, and brushes, while the red plastic cap at the back holds the electrical terminals and provides structural support. The output shaft protrudes from the front and can be connected to gears, pulleys, or propellers for mechanical motion. Due to its compact size, lightweight construction, and high-speed operation, this mini motor is ideal for battery-powered applications requiring simple, low-cost motion control. However, since it relies on brushes for commutation, it experiences wear over time, leading to reduced lifespan compared to brushless motors.

Fig.3.7 Mini DC Motor

3.3.7 Limit Switch

A limit switch is an electromechanical device that detects the presence, position, or movement of an object by making or breaking an electrical connection when a mechanical actuator is engaged. It consists of a durable plastic or metal housing, a lever actuator, and three terminals: Common (C), Normally Open (NO), and Normally Closed (NC), allowing it to function in various control circuits. The switch operates on a snap-action mechanism, meaning a small movement of the actuator results in a rapid change of state, ensuring precise and reliable switching. Limit switches are widely used in industrial automation, machinery, robotics, and conveyor systems to detect end positions, prevent over-travel, and provide safety interlocks.

Fig.3.8 Limit Switch

3.4 SOFTWARE USED

Arduino IDE: To write and upload programs to Arduino compatible board.

3.4.1 Arduino IDE

The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino hardware to upload programs and communicate with them.

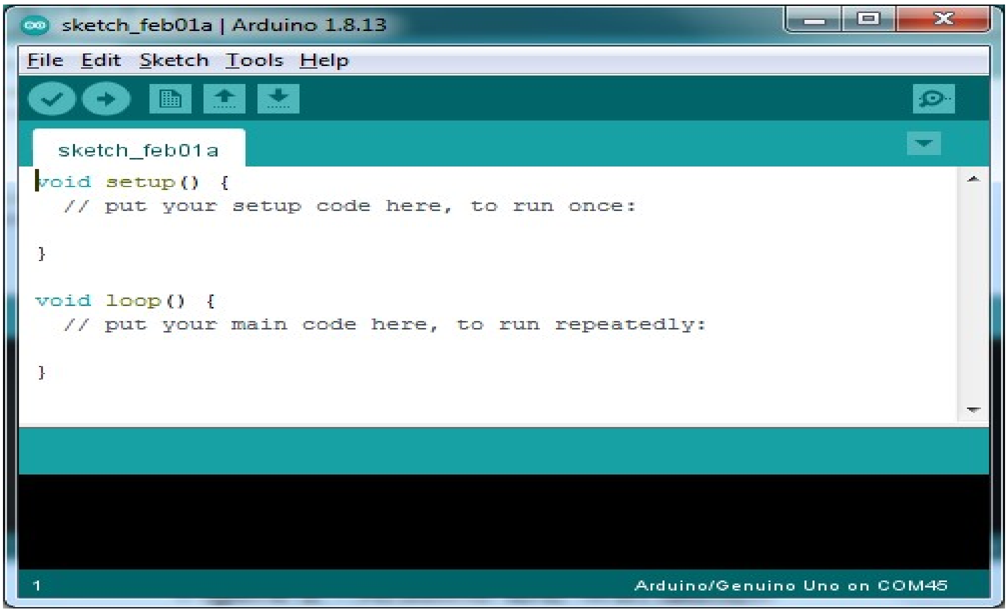

Fig.3.9 Model of Programme of Arduino IDE

Programs written using Arduino Software (IDE) are called sketches. These sketches are written in the text editor and are saved with the file extension. The editor has features for cutting/pasting and for searching/replacing text. The message area gives feedback while saving and exporting and also displays errors. The console displays text output by the Arduino Software (IDE), including complete error messages and other information. Versions of the Arduino Software (IDE) prior to 1.0 saved sketches with the extension. It is possible to open these files with version 1.0, you will be prompted to save the sketch with the extension onsave.

CHAPTER 4

DESIGN AND FABRICATION

This chapter deals with the Design and fabrication of automatic bike side stand slider.

4.1 CIRCUIT DIAGRAM

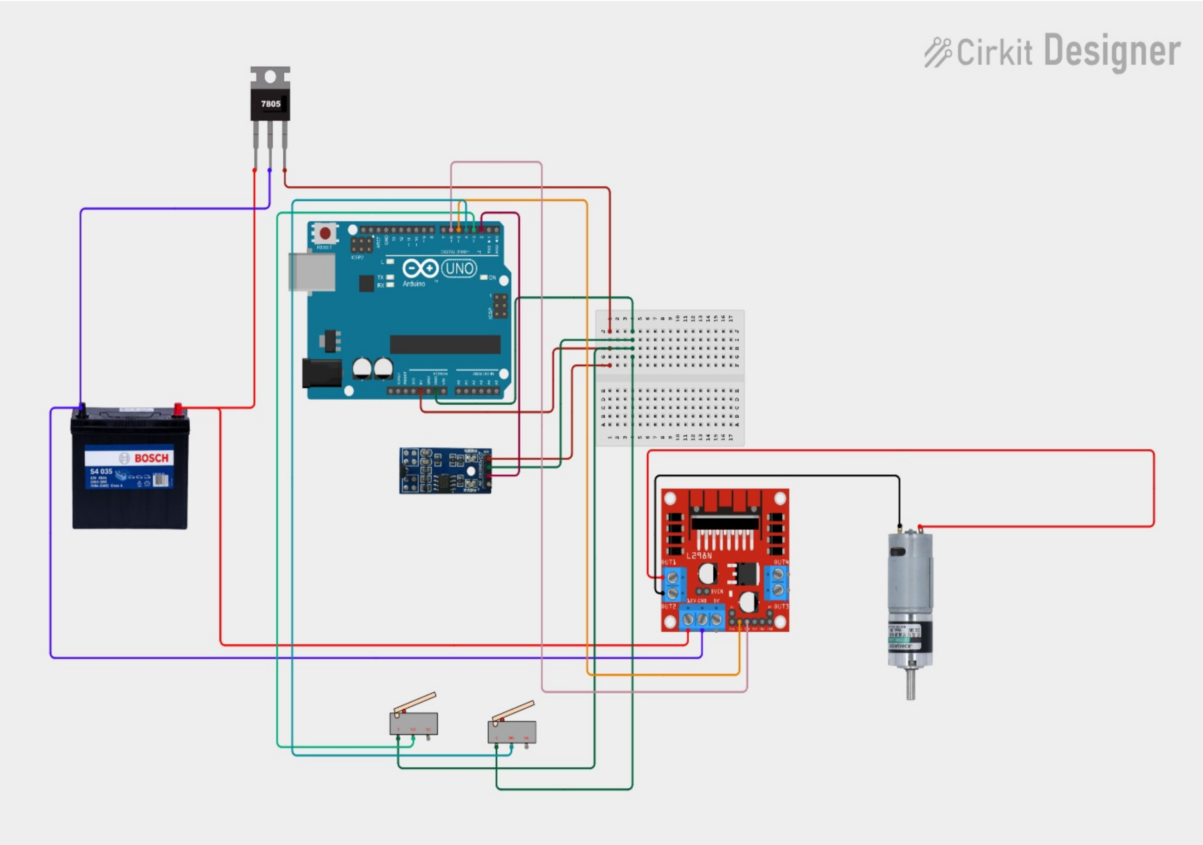

Fig.4.1 Circuit Diagram

CHAPTER 5

TOTAL COST

SI No	PRODUCT	QUANTITY	PRICE
1	Arduino UNO R3	1	499
2	Voltage Regulator	1	9
3	Brushed DC Motor	1	225
4	Motor Control Drive	1	150
5	Hall Effect Speed Sensor	1	89
6	Mini DC Motor	1	15
7	Limit Switch	2	48
8	BreadBoard	1	70
9	Jumper wires	2	150
		TOTAL	**1,255**

Table 5.1 Total Cost

CHAPTER 6

IMPLEMENTATION AND RESULT

6.1 IMPLEMENTATION

This project consist of various elements like Arduino, Hall effect sensor, Brushed DC motor, Motor control drive, Battery, Voltage regulator.

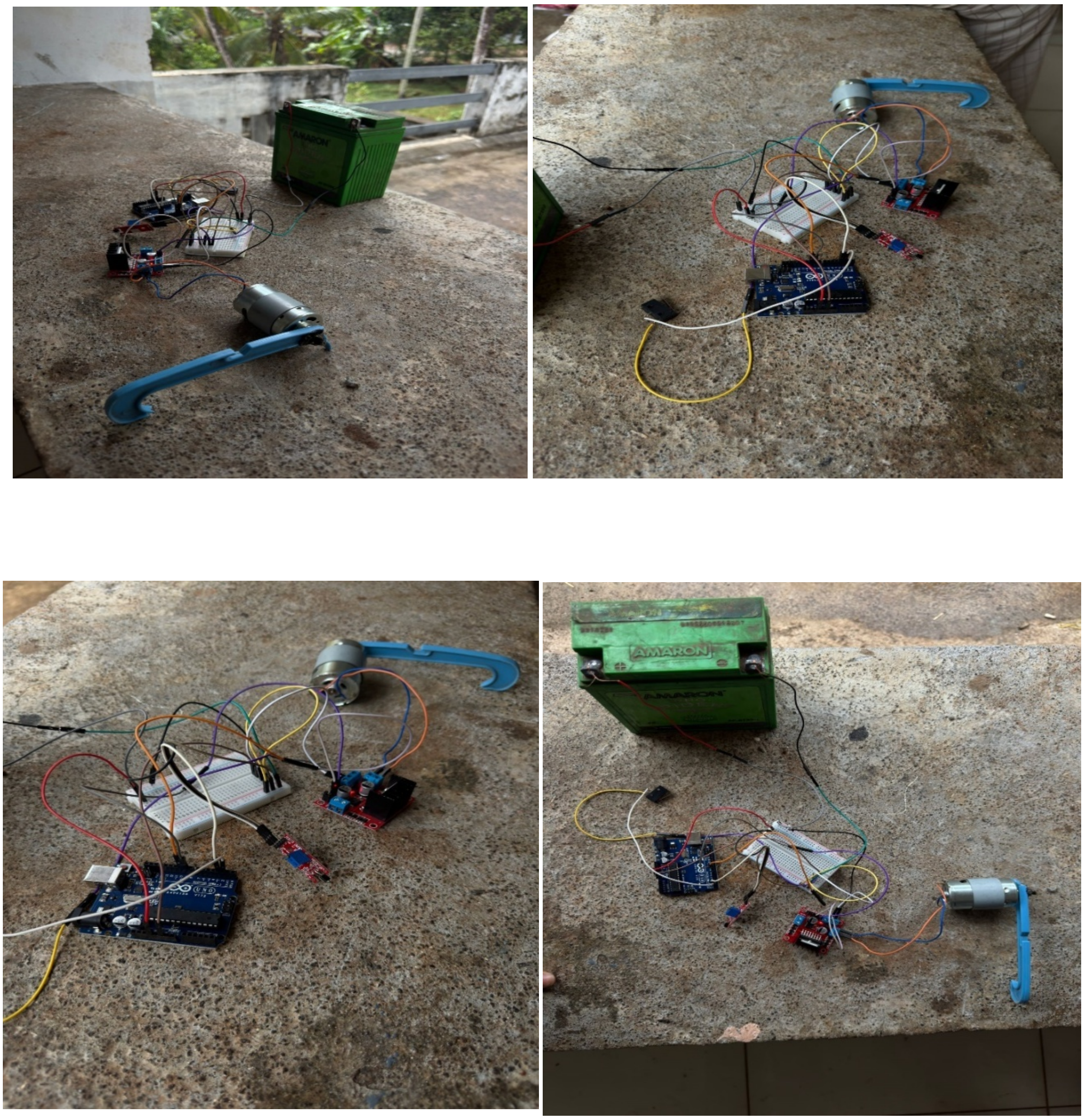

Fig.6.1 Implementation of components

6.2 RESULT

The implementation of the Automatic Motorcycle Side Stand Retraction System has yielded promising results in terms of enhancing rider safety, improving operational convenience, and ensuring seamless integration with existing motorcycle components. Through extensive testing, the system has demonstrated its ability to effectively detect the motorcycle's speed using the Hall-effect speed sensor, process the data through the Arduino microcontroller, and activate the Brushed DC motor via the L298N motor driver to retract the side stand once the speed exceeds 10 km/h, thereby eliminating the risk of riders accidentally riding with an extended side stand, which could otherwise result in severe accidents, particularly during turns or uneven terrain. Additionally, the incorporation of two limit switches has proven to be an essential feature, ensuring precise control over the stand's positioning by stopping the motor when the stand reaches either the fully extended or fully retracted position, preventing excessive force application, mechanical strain, and system malfunctions.

Furthermore, the system's reliance on the motorcycle's 12V battery for power supply has confirmed its practicality, as it allows for continuous and reliable operation without the need for additional power sources, thereby maintaining energy efficiency while ensuring smooth performance. The use of a Brushed DC motor, coupled with the L298N motor driver, has provided a cost-effective yet highly efficient mechanism for actuating the side stand, demonstrating optimal torque and speed characteristics necessary for safe and reliable operation. The project has also shown that the system can be installed with minimal modifications to the motorcycle's structure, making it adaptable to a wide range of motorcycle models without requiring complex mechanical alterations, thus increasing its feasibility for widespread adoption.

Overall, the results of this project indicate that the Automatic Bike Side Stand Slider effectively addresses a significant safety concern among motorcyclists by preventing avoidable accidents caused by an extended side stand, while also offering a cost-efficient, reliable, and user-friendly solution that enhances both safety and riding convenience, making it a valuable innovation in the field of two-wheeler safety technology.

CHAPTER 7

FUTURE SCOPE

The future scope of the Automatic Motorcycle Side Stand Retraction System extends across multiple domains, including enhanced safety features, integration with advanced sensor technology, improved automation, and broader adaptability to different types of motorcycles, making it a promising innovation in the field of two-wheeler safety and automation. One of the most significant future advancements could involve the incorporation of wireless connectivity and IoT (Internet of Things) integration, allowing real-time monitoring of the system's status through a mobile application, which would enable riders to receive alerts or notifications regarding the stand's position, battery status, and system performance, thereby enhancing user awareness and control.

Furthermore, the system can be improved by integrating additional sensors, such as an inertial measurement unit (IMU) or tilt sensors, to enhance accuracy in detecting the motorcycle's movement and lean angle, which would allow the stand to retract not only based on speed but also when the bike begins moving forward or tilts significantly, thus increasing its reliability and responsiveness in various riding conditions. Another potential improvement could involve the implementation of a brushless DC motor (BLDC) or a stepper motor in future iterations, as these motors offer higher efficiency, longer lifespan, and precise control, which could further enhance the durability and performance of the system.

Additionally, future versions of this system could incorporate self-diagnostic capabilities and fault detection mechanisms, where the microcontroller continuously monitors the functioning of components, such as the motor, limit switches, and power supply, and provides troubleshooting feedback or error alerts in case of malfunctions, reducing the likelihood of system failures and improving long-term reliability. The design can also be optimized for compatibility with electric motorcycles and scooters, ensuring that the system aligns with the growing trend of electric mobility, where energy efficiency and smart automation are key priorities.

Moreover, as motorcycles become increasingly smart and connected, there is potential for integration with advanced rider assistance systems (ARAS), where the side stand retraction mechanism could work in conjunction with other safety features such as ABS (Anti-lock Braking System), traction control, and collision avoidance systems to enhance overall rider protection. Future developments could also explore voice or gesture-based activation for the side stand mechanism, enabling riders to control the stand using voice commands or simple gestures, making it even more intuitive and user-friendly.

Lastly, expanding the project's commercial viability and market adoption through collaboration with motorcycle manufacturers and aftermarket accessory companies could lead to large-scale production and adoption, ensuring that both new and existing motorcycle models can benefit from this safety innovation. By continuously refining the design, optimizing the cost, and expanding its functional capabilities, this project has the potential to become a standard safety feature in motorcycles worldwide, significantly reducing accident risks and improving the overall riding experience for motorcyclists.

CHAPTER 8

CONCLUTION

The Automatic Bike Side Stand Slider is an innovative solution designed to enhance rider safety and convenience by automatically retracting the side stand when the motorcycle reaches a speed of 10 km/h. By integrating Arduino-based control, a Hall effect speed sensor, and a motor-driven mechanism, the system eliminates the risk of accidents caused by a forgotten side stand. The use of limit switches and a motor driver circuit ensures precise and reliable operation. This project not only addresses a common safety hazard but also paves the way for future advancements such as IoT integration, AI-driven automation, and adaptive designs for different motorcycles. With further refinements, this technology has the potential to become a standard safety feature in modern motorcycles, significantly reducing accidents and improving the overall riding experience.

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APPENDIX

```
#define IN1 9           // Motor Driver Input 1

#define IN2 10          // Motor Driver Input 2

#define LIMIT_SWITCH 2   // Limit Switch Pin

#define HALL_SENSOR 3    // Hall Effect Sensor Pin



bool moveForwardNext = true; // Track motor direction for normal operation

bool buttonPressed = false;  // To prevent multiple triggers on limit switch

int hallCount = 0;           // Counter for hall sensor detections

bool hallTriggered = false;  // For debouncing the hall sensor



void setup() {

pinMode(IN1, OUTPUT);

pinMode(IN2, OUTPUT);

pinMode(LIMIT_SWITCH, INPUT_PULLUP);

pinMode(HALL_SENSOR, INPUT_PULLUP);  // Configure as needed for your sensor

Serial.begin(9600);

}



void loop() {

  // --- Limit Switch Based Operation ---

  if (digitalRead(LIMIT_SWITCH) == LOW) { // Limit switch pressed

    if (!buttonPressed) {  // Ensure single trigger per press
```

```
buttonPressed = true;

Serial.println("Limit switch activated:");

            if (moveForwardNext) {

moveForward();

            } else {

moveBackward();

            }

delay(1000); // Run motor for 1 second

stopMotor();

moveForwardNext= !moveForwardNext; // Toggle direction

        }

    } else {

buttonPressed = false; // Reset button state when released

    }


    // --- Hall Sensor Based Operation ---

    if (digitalRead(HALL_SENSOR) == LOW) { // Hall sensor triggered (assumes LOW when magnet is near)

        if (!hallTriggered) {  // Debounce so we count one event per detection

hallTriggered = true;

hallCount++;

Serial.print("Hall sensor count: ");

Serial.println(hallCount);
```

```
delay(50);  // Small debounce delay
        }
    } else {
hallTriggered = false;
    }

    // Check if hall sensor has been triggered 10 times
    if (hallCount>= 10) {
Serial.println("10 hall sensor detections reached: Rotating motor backward for 5 seconds.");
moveBackward();
delay(5000); // Rotate backward for 5 seconds
stopMotor();
hallCount = 0; // Reset hall count after special action
    }
}

void moveForward() {
digitalWrite(IN1, HIGH);
digitalWrite(IN2, LOW);
Serial.println("Motor moving forward.");
}

void moveBackward() {
```

```
digitalWrite(IN1, LOW);

digitalWrite(IN2, HIGH);

Serial.println("Motor moving backward.");

}


void stopMotor() {

digitalWrite(IN1, LOW);

digitalWrite(IN2, LOW);

Serial.println("Motor stopped.");

}
```

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```
#define IN1 9           // Motor Driver Input 1

#define IN2 10          // Motor Driver Input 2

#define LIMIT_SWITCH 2   // Limit Switch Pin

#define HALL_SENSOR 3    // Hall Effect Sensor Pin


bool moveForwardNext = true; // Track motor direction for normal operation

bool buttonPressed = false;  // To prevent multiple triggers on limit switch

int hallCount = 0;           // Counter for hall sensor detections

bool hallTriggered = false;  // For debouncing the hall sensor


void setup() {
```

```
pinMode(IN1, OUTPUT);
pinMode(IN2, OUTPUT);
pinMode(LIMIT_SWITCH, INPUT_PULLUP);
pinMode(HALL_SENSOR, INPUT_PULLUP);  // Configure as needed for your sensor
Serial.begin(9600);
}


void loop() {
  // --- Limit Switch Based Operation ---
  if (digitalRead(LIMIT_SWITCH) == LOW) { // Limit switch pressed
    if (!buttonPressed) {  // Ensure single trigger per press
buttonPressed = true;
Serial.println("Limit switch activated:");
      if (moveForwardNext) {
moveForward();
      } else {
moveBackward();
      }
delay(1000); // Run motor for 1 second
stopMotor();
moveForwardNext= !moveForwardNext; // Toggle direction
    }
  } else {
```

```
buttonPressed = false; // Reset button state when released
    }

    // --- Hall Sensor Based Operation ---
    if (digitalRead(HALL_SENSOR) == LOW) { // Hall sensor triggered (assumes LOW when magnet is near)
        if (!hallTriggered) {  // Debounce so we count one event per detection
hallTriggered = true;
hallCount++;
Serial.print("Hall sensor count:5 ");
Serial.println(hallCount);
delay(50);  // Small debounce delay
        }
    } else {
hallTriggered = false;
    }

    // Check if hall sensor has been triggered 10 times
    if (hallCount>= 5) {
Serial.println("5 hall sensor detections reached: Rotating motor backward for 5 seconds.");
moveBackward();
delay(1000); // Rotate backward for 5 seconds
stopMotor();
```

```
hallCount = 0; // Reset hall count after special action
    }
}

void moveForward() {
digitalWrite(IN1, HIGH);
digitalWrite(IN2, LOW);
Serial.println("Motor moving forward.");
}

void moveBackward() {
digitalWrite(IN1, LOW);
digitalWrite(IN2, HIGH);
Serial.println("Motor moving backward.");
}

void stopMotor() {
digitalWrite(IN1, LOW);
digitalWrite(IN2, LOW);
Serial.println("Motor stopped.");
}
```